

Source-free domain adaptation framework for fault diagnosis of rotation machinery under data privacy

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ABSTRACT

Domain adaptation-based fault diagnosis method can learn knowledge from the labeled data in the related source rotation machinery and achieve reliable performance in the unlabeled target machinery. However, due to the impact of data privacy, the labeled source data are usually unavailable, severely restricting the application of the domain adaptation-based fault diagnosis methods. In this paper, a source-free domain adaptation framework is proposed to decouple the direct usage of the source data. First, the adaptive prototype memory matrix is constructed to select reliable samples, which are employed to define the pseudo labels. Then, the confidence-based filter is designed to improve the reliability of pseudo labels. Finally, the curriculum learning strategy is utilized to balance the optimal weight between the source information and target data. The effectiveness of the proposed method was verified by datasets in the rig and real wind turbine gearboxes.

1. Introduction

Rotation machinery plays a significant role in industrial manufacturing, such as wind turbines, aero engines, and gas turbines [1, 2]. However, due to the long-time alternate loads and harsh operating environment, the critical components of rotation machinery inevitably encounter performance degradation and even faults, which may cause high maintenance costs and safety disasters. Accurate fault diagnosis can help to eliminate potential hazards and reduce the rate of equipment failure, and more importantly can avoid safety accidents [3]. Therefore, it plays a critical role in safeguarding production safety and enhancing equipment reliability. [4,5].

Deep learning [6], owing to the automatic feature learning ability and nonlinear representation capabilities, can directly learn useful fault features from monitoring data of mechanical equipment [7]. In recent years, deep learning methods, such as convolutional neural networks (CNN) [8,9], long short-term memory [10], and graph networks [11, 12], have been widely applied in the fault diagnosis for rotation machinery, which can help avoid major accidents and improve production safety and reliability by accurately identifying potential faults [13,14]. The success of the above methods is based on two assumptions: one is sufficient labeled training data, and the other is that the training and test data are independent and identically distributed [15,16]. However, the labeled fault samples are usually rare and even unavailable due to the

uncertainty of fault occurrence and the long-term regular operation of gearboxes. Furthermore, there are considerable distribution discrepancies between the monitoring signals collected by different gearboxes and even different sensors, which does not meet the assumption of independent and identical distribution.

Domain adaptation can effectively alleviate the problem of insufficient labeled data and data distribution discrepancy [17,18]. The main idea of the domain adaptation method is to utilize the relevant source domain data to train and apply it to the target domain task, which aims to learn domain-invariant features from the source domain and target domain [19,20]. In recent years, domain adaptation methods have received extensive attention in fault diagnosis [21]. For instance, Qian et al. [22] combined maximum mean discrepancy (MMD) and Correlation alignment (CORAL) to further enhance the distribution alignment, which achieved higher performance than the typical domain adaptation methods. Shi et al. [23] designed a transferable adaptive channel attention module to enhance the positive transfer, which can effectively tackle the negative transfer. To further improve the diagnosis accuracy, some scholars tried to employ knowledge from multiple source domains. Li and Yu [24] proposed a feature-lever and class-lever multi-source domain adaptation method, which can improve the diagnosis performance under varying working conditions. Li et al. [25] proposed a cross-attribute domain adaptation network to learn global and local information from multiple source domains with different sampling

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frequencies. Yang et al. [26] proposed partial distribution adaptation sub-networks to learn features from multiple source domains, and then the decisions from multiple sub-networks were combined to identify unseen faults of target machines. Ding et al. [27] proposed a deep imbalanced domain adaptation framework to solve the problem about feature shift and label shift under different working conditions. The above research effectively improves the fault diagnosis accuracy of the model in the target domain task by learning the source domain data characteristics of different data distributions and enhances the equipment reliability and system security.

By the above literature review, current domain-adaptation-based fault diagnosis methods need to ensure that source domain data is available during the training stage of the model. However, source domain data is usually unavailable in practical applications of rotation machinery. The main reason is that the monitoring data of rotation machinery, such as wind turbines, aero engines, and gas turbines, usually contains characteristics of the structure and confidential information, as well as the fault labels of rotation machinery involving potentially valuable commercial information. With increasing concerns about data privacy, enterprises and companies will not share the fault data, but the pre-trained model. Consequently, the above domain adaptation-based fault diagnosis methods will be severely restricted or even not applicable in the case of data privacy.

To address the problem above, a source-free domain adaptation framework (SFDAF) is proposed in this paper for fault diagnosis of rotation machinery. The main idea of the proposed method is to decouple the domain adaptation process from the direct usage of source rotation machinery data by leveraging the pre-trained source model. First, the self-entropy of every target sample can be obtained through the pre-trained source model, and the target samples with lower self-entropy are selected as reliable samples. Next, the entropy-based adaptive prototype memory matrix (EAPMM) can be formed with the features of reliable samples, and pseudo labels of target samples can be defined according to the similarity between the features of target samples and the EAPMM. Then, a confidence-based pseudo-label filter is constructed by Hausdorff distance to filter for pseudo-labels, and the confidence of pseudo-labels can be further improved. Finally, the curriculum learning strategy is utilized to balance the optimization weights of the target model between the pre-trained source model and the pseudo-labels, which help the target model learn invariant features from the source model from “easy” to “hard.” The main contributions of this literature are summarized as follows.

- (1) The source-free domain adaptation framework is proposed to solve the data privacy problem in fault diagnosis for rotation machinery, which can decouple the domain adaptation from labeled source data.
- (2) An adaptive entropy thresholds method is proposed without additional hyperparameters, which construct the entropy-based adaptive prototype memory matrix to effectively build reliable sample prototypes and obtain pseudo-labels for the target domain.
- (3) Confidence-based filter is designed with Hausdorff distance, which can improve the reliability of the pseudo-labels in the target domain by filtering out the false labels.
- (4) A training strategy based on curriculum learning is proposed to help gradually transfer the focus of the pre-trained model from retaining source domain information to learning target features, thereby further learning the domain-invariant features.

The remainder of this article is organized as below. The proposed source-free domain adaptation framework is described in Section 2. Then, Section 3 details the fault diagnosis procedure of the proposed method. In Section 4, experimental evaluations are carried out to demonstrate the effectiveness of the proposed method. Finally, the main conclusions are drawn in Section 5.

2. Source-free domain adaptation framework

2.1. Problem setup

This paper focuses on source-free domain adaptation for fault diagnosis of rotation machinery. The main distinction between UDA and the proposed SFDAF is that the training data of SFDAF only contains unlabeled target data without any source samples. In general, this study is carried out under the following definitions and assumptions:

Given a vibration dataset $\{x_i, y_i | x_i \in X, y_i \in Y\}_{i=1}^N$, where x_i denotes the vibration signal and y_i indicates the health state of the vibration signal, the data space X obeys the marginal probability distribution $p(X)$, and the label space $Y = \{1, 2, \dots, k\}$ includes k kinds of health states. The source domain $D_s = \{(x_i^s, y_i^s) | x_i^s \in X^s, y_i^s \in Y^s\}_{i=1}^{N_s}$ consists of a labeled vibration dataset, and the target domain $D_t = \{(x_i^t) | x_i^t \in X^t\}_{i=1}^{N_t}$ is composed of an unlabeled vibration dataset. Specifically, the probability distribution of the source domain is related to that of the target domain. The main goal of domain adaptation is to minimize the discrepancy between the source and target domain.

In the source-free domain adaptation, due to the limitation of the data privacy problem, the source domain data is unavailable, while the parameters of the pre-trained model θ_s from the source domain can be shared. More precisely, source-free domain adaptation aims to achieve the domain adaptation with the unlabeled target domain $D_t = \{(x_i^t) | x_i^t \in X^t\}_{i=1}^{N_t}$ and the parameters of the pre-trained model θ_s .

2.2. Source-free domain adaptation framework

The overall flow of the proposed source-free domain adaptation framework is illustrated in Fig. 1. The SFDAF mainly contains two branches: a frozen pre-trained source model and a trainable target model. The pre-trained source model consists of a feature extractor F_s and a classifier C_s , and the parameters of the source model are fixed after being pre-trained with the source data. The trainable target model consists of a feature extractor F_t and two classifiers, C_{s2t} and C_t . The initial parameters of the target model are shared by the pre-trained source model. In this paper, the feature extractors of the two models consists of three convolutional layers, three batch normalization layers and three max-pooling layers, the classifiers of the source and target models consists of three fully connected layers.

In the training stage, target data are fed into the pre-trained source model and the target model, respectively. The pre-trained source model can obtain the source pseudo label $\bar{y}_s$ that indicates the predicted results under the source information. The predicted results from the C_{s2t} are calculated and combined with the $\bar{y}_s$ to calculate the source-oriented loss L_{src} under the constrain of the source information. The source-oriented loss function aims to maintain the information of the source domain. In the other branch C_t , the predicted results of the target data are calculated, and then the similar score is calculated according to the EAPMM to define the target pseudo label $\bar{y}_t$. Then, combined the confidence score that obtained by the confidence-based filter, the target-oriented loss L_{tar} is constructed. Finally, the parameters of the proposed SFDAF are optimized through the backpropagation.

2.3. Entropy-based adaptive prototype memory matrix

Information entropy is used to describe the uncertainty of the occurrence of possible events, where the low entropy with the low uncertainty in the probability event and the high entropy with the high uncertainty of the probability event. In the deep learning-based fault diagnosis model, the output of the last layer is usually used as a classifier to judge the health status of the mechanical equipment. Besides, the results of the output layer can also represent the uncertainty of the model prediction. The main idea of pseudo-labels based on entropy has

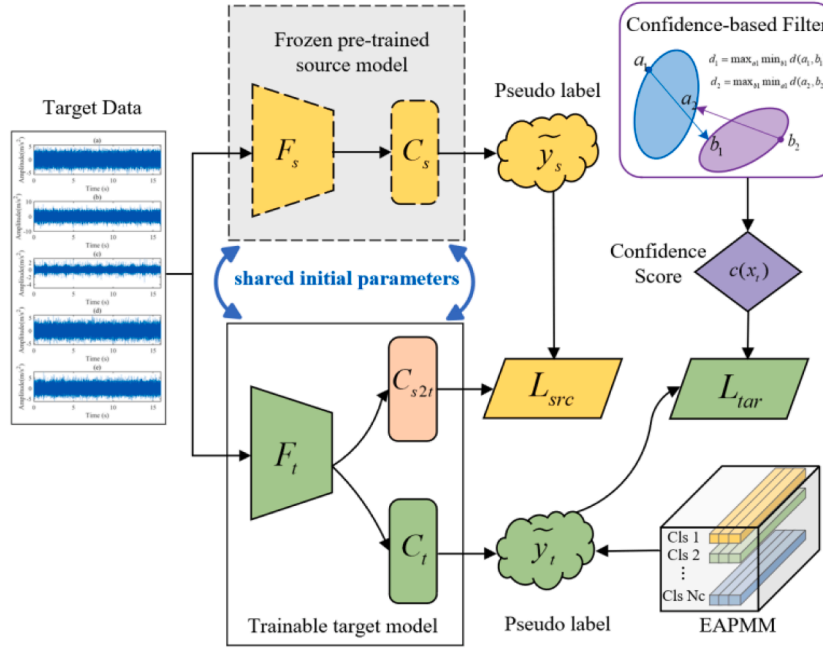


Fig. 1. Overall flow of the proposed source-free domain adaptation framework.

been widely used in the field of fault diagnosis [28,29]. Inspired by this, this paper utilized entropy for pseudo-label calculation of vibration signals. Thus, self-entropy is applied in this paper to describe the uncertainty of the prediction. The calculation formula of the self-entropy is expressed by

$$H(X) = -\sum_{x \in X} p(x) \log p(x) = E \left(\log \frac{1}{p(X)} \right) \quad (1)$$

where x indicates the random variables, X represents the collection of random variables, $p(x)$ represents the function of probability.

Domain adaptation fault diagnosis aims to learn fault characteristics from the related source domain to solve the diagnosis problem in the target domain. Based on the assumption that the source and target domains are close for the same fault status, the predicted probability distribution of the target domain can be obtained by the pre-trained source fault diagnosis model. Although the working condition between the source and target domain is different, the predicted diagnosis results of target vibration data with lower self-entropy are reliable. On the contrary, samples with high entropy have high uncertainty in prediction with the pre-trained source model. Therefore, reliable target samples are selected with the principle of low self-entropy in each class, and the self-entropy of reliable samples is normalized by

$$H_{n_self}(x_t) = -\frac{1}{\log N_c} \sum l(x_t) \log(l(x_t)) \quad (2)$$

where x_t denotes the target sample, $l(x_t)$ denotes the predicted probability of the target data, N_c represents the class number.

Then, class-wise entropy set in target domain H_{set} can be obtained by

$$H_{set} = \{H_1, \dots, H_{N_c}\} \quad (3)$$

$$H_c = \{H_{n_self}(x_t) | x_t \in X_c\}$$

where H_c indicates entropy set of class c data in the target domain, X_c refers to the target data in class c .

Although the source and target domain are related and similar, there still exist distribution discrepancies that lead to different entropy distributions. Specifically, due to the marginal and conditional probability discrepancies, the entropy value of certain categories in the target domain may be lower or higher than that of other categories. Further-

more, due to the discrepancies between the source domain and the target domain, there are some problems exist in samples with low entropy values such as low quality and unreliability, which means that even if a sample has a low entropy value, it still does not belong to the predicted category by the pre-trained model. Regarding the issue above, this paper proposed an approach that can adaptively select the appropriate number of reliable target samples for each category. The main idea is to set the maximum value of the minimum entropy value of each category as the adaptive threshold.

$$\eta = \max\{\min(H_c) | c \in C\} \quad (4)$$

where η represents the threshold of the entropy value and C indicates the class set of the target domain. Then, reliable target samples in each class can be selected with the adaptive threshold, and features of the reliable samples obtained by the pre-trained source model are collected as the prototypes, which the entropy-based adaptive prototype memory matrix is constructed by

$$P_{set} = \{P_1, \dots, P_{N_c}\} \quad (5)$$

$$P_c = \{F_t(x_t) | x_t \in X_c, H(x_t) \leq \eta\}$$

where F_t represents the trainable feature extractor in target domain, P_c indicates the prototypes set of class c . The proposed EAPMM can guarantee as least one reliable target samples in each class without any additional hyperparameters.

Then the features of target samples can be obtained by the pre-trained source model, and the pseudo label method [29] are applied to represent the unlabeled data, which defined by the similarity score S_c between the constructed EAPMM and the extracted features.

$$S_c(x_t) = \frac{1}{|P_c|} \sum_{m_c \in P_c} \frac{m_c^T F_t(x_t)}{\|m_c\|_2 \|F_t(x_t)\|_2} \quad (6)$$

where m_c indicates one of the prototypes in the class c . Due to the different number multiple prototypes in each class, the pseudo label is defined as the class that obtain the best similar score.

$$\bar{y}_t = \operatorname{argmax}\{S_c(x_t) | c \in C\} \quad (7)$$

where the $\bar{y}_t$ indicates the pseudo label.

2.4. Confidence-based filter

With the pseudo labels obtained by the EAPMM, the conventional cross-entropy loss function is usually applied to optimize the parameters of the model. However, due to the probability distribution discrepancies between the source and the target domain and no ground truth labels used, the defined pseudo labels are not reliable enough, and the false pseudo labels may lead to the error propagation to the model optimization, which is one of the main reasons for poor performance in model training, affects the classification accuracy of the model [30]. To address the problem above, this paper applied a confidence-based filter to further improve the reliability of the pseudo labels. The main idea is to estimate the confidence of pseudo labels by the Hausdorff distance between the features of target samples and the EAPMM in the high dimensional space, which can consider the influence of each element by the set-to-set distance. This may drop out some samples with pseudo labels that really true, which prevents the target model from learning these data features. But it can relatively filter out false pseudo labels and reduce the misleading of false labels in the process of backpropagation to model parameter optimization.

Comprehensive consideration of computational complexity, the confidence-based filter mainly compares the distance between the two most similar prototype sets and the target domain features and then calculate the confidence. Specifically, given a target sample x_t , the singleton set $Q = \{F_t(x_t)\}$ can be obtained by the feature extractor of target domain, and then the similarity score can be calculated with the EAPMM, where the class prototypes set P_{t1} with highest similar score is defined as the pseudo label of the target sample, and the class prototypes set P_{t2} second similar score is neglected. In the confidence-based filter, these two prototypes are considered to estimate the Hausdorff distance. The Hausdorff distance between P_{t1} and the Q in the metric space is given by

$$d_H(Q, P_{t1}) = \max \left\{ \sup_{q \in Q} \inf_{p \in P_{t1}} d(q, p), \sup_{p \in P_{t1}} \inf_{q \in Q} d(q, p) \right\} \quad (8)$$

where p and q represent one of the prototype P_{t1} and the Q , $d(q, p)$ is the distance metric function measured by

$$d(q, p) = 1 - \frac{q \cdot p}{\|q\|_2 \|p\|_2} \quad (9)$$

The sets Q and P_{t1} are closed and discretized, therefore, the suprema and infima can be converted as maximum and minimum respectively.

$$\begin{aligned} d_H(Q, P_{t1}) &= \max \left\{ \max_{q \in Q} \min_{p \in P_{t1}} d(q, p), \max_{p \in P_{t1}} \min_{q \in Q} d(q, p) \right\} \\ &= \max \left\{ \min_{p \in P_{t1}} (F_t(x_t), p), \max_{p \in P_{t1}} (F_t(x_t), p) \right\} \\ &= \max_{p \in P_{t1}} (F_t(x_t), p) \end{aligned} \quad (10)$$

Then the maximum distance between Q and P_{t1} in the metric space can be obtained. For the second most similar prototypes sets P_{t2} , the distance function is modified to obtain the minimum deviation between Q and P_{t2} .

$$\begin{aligned} \widehat{d}_H(Q, P_{t2}) &= \min \left\{ \sup_{q \in Q} \inf_{p \in P_{t2}} d(q, p), \sup_{p \in P_{t2}} \inf_{q \in Q} d(q, p) \right\} \\ &= \min_{p \in P_{t2}} (F_t(x_t), p) \end{aligned} \quad (11)$$

In the confidence-based filter, the confidence score can be obtained by comparing the value of the two distances, which represents the reliability of the pseudo labels. The definition criterion of confidence score is that only when the maximum deviation value of the most similar class is less than the minimum deviation value of the second similar class, the given target sample is defined as a reliable sample. The schematic diagram of the confidence score is shown in Fig. 2. The confidence score is defined as follows.

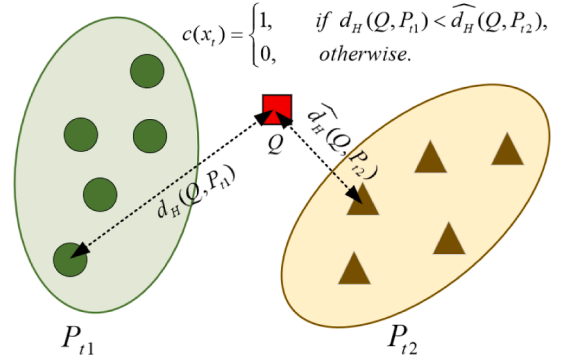


Fig. 2. Schematic diagram of confidence score.

$$c(x_t) = \begin{cases} 1, & \text{if } d_H(Q, P_{t1}) < \widehat{d}_H(Q, P_{t2}), \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

2.5. Curriculum learning strategy and optimization

The optimization object mainly including two parts: the source-regularization loss and the self-learning loss.

The source-regularization loss mainly used to retain feature information of source domain and constrain the self-learning of the target domain data, which can prevent the problem of self-bias. The conventional cross-entropy loss is applied to regularize the predicted pseudo labels of the pre-trained model. The calculation method is as follows.

$$L_{src}(D_t) = -\mathbb{E}_{x_t \sim D_t} \sum_{c=1}^{N_c} p_s(x_t) \log(\sigma(C_{s2t}(F_t(x_t)))) \quad (13)$$

where $p_s(x_t)$ denotes the estimated probability with the pseudo label $\bar{y}_s$ predicted by the pre-trained model, $\sigma(\cdot)$ indicates the SoftMax function.

The object of the self-learning loss function is the predicted results of the target data and its corresponding pseudo-label, which to ensure the model learns the feature distribution information of the target domain through self-learning. The self-learning loss function can be formulated as follows:

$$L_{tar}(D_t) = -\mathbb{E}_{x_t \sim D_t} \sum_{c=1}^{N_c} c(x_t) p_t(x_t) \log(\sigma(C_t(F_t(x_t)))) \quad (14)$$

Where $p_t(x_t)$ denotes the estimated probability with the pseudo label $\bar{y}_t$.

During the training process of the model, the self-learning of the target domain is constrained by the features of the source domain. Since the parameter optimization direction of the model has different sensitivity to the above two losses, the weights of the two losses are directly affect the performance of the model. In the domain adaptation procedure of the pre-trained source model, the vibration data of the target domain is trained in batches, which cannot fully represent the real data distribution of the target domain. What's more, the significant discrepancies between the source and target domains lead to the fact that the first several batches of vibration data in the target domain can easily guide the pre-trained model to the wrong diagnostic decision boundary. In this stage, pseudo labels are not reliable enough, the source domain information should be focused to avoid error propagation direction led by false pseudo labels. Therefore, in the early stage of training, it is necessary to keep the constraints of the source domain information on the model as much as possible. As the number of model training epochs increases, the distribution of the target domain data is gradually learned, and the reliability of pseudo-labels gradually increases. Therefore, in the later stage of model training, the learning attention of the model is transferred to the feature information of the target domain, so that the model can be applied to the target domain task.

In this paper, A training strategy based on the curriculum learning strategy [31] is proposed to shift the learning focus between the source-regularization loss and the self-learning loss, which follows an “easy-to-hard” scheme and gradually transfer the focus of the pre-trained model from retaining source domain information to learning target features. Therefore, the total loss function of the SFDAF with the curriculum learning strategy is formulated as:

$$L_{total}(D_t) = (1 - \alpha)L_{src}(D_t) + \alpha L_{tar}(D_t) \quad (15)$$

where α is the dynamic weight of the source-regularization loss and the self-learning loss, which decided by curriculum learning strategy with a progressive schedule [32]:

$$\alpha = \frac{2}{1 + e^{-10 \times \frac{T}{T_{max}}}} \quad (16)$$

where the T presents the current epoch and the T_{max} indicates the number of total training epochs.

Fig. 3 shows the dynamic weight α setting with the change of iteration number of the training stage, where the loss weight α is setting as 0 from the first epoch and gradually increases to 1 at the end of the epoch. In summary, the complete process of the proposed SFDAF is tabulated in Algorithm 1.

2.6. Fault diagnosis procedure of the proposed SFDAF

In this study, a SFDAF method for fault diagnosis of rotation machinery is developed. The fault diagnosis flowchart based on the proposed SFDAF is illustrated in Fig. 4. The general process mainly includes four parts: Data acquisition, Model training, and Diagnosis and Analysis.

- (1) Data acquisition: First, the accelerometers are applied to collect the raw vibration data of the rotation machinery, such as wind turbine gearboxes. Then, the collected signal is preprocessed, specifically, the sliding window operation is performed on the collected raw vibration data to generate samples with a fixed number of points, which are randomly selected and divided into training data set and test data.
- (2) Model Training: First, analyze the fixed pre-trained source model, the trainable target model is constructed with the same structure as the pre-trained source model, and initialized shared parameters. Then, the training dataset is fed into the pre-trained source model and target model, respectively, the outputs from the source model C_s and target model C_{s2t} are obtained, which are utilized to calculate source regularization loss, while the output from the target model C_t is applied to construct EAPMM, which can select reliable samples to help define the pseudo label. Next, the confidence-based filter is designed to improve the quality of the pseudo label, and then calculate the self-learning loss. Finally, the curriculum learning strategy is employed to weigh the two losses and optimized parameters by backpropagation algorithms. The optimal target model parameters are obtained and saved.

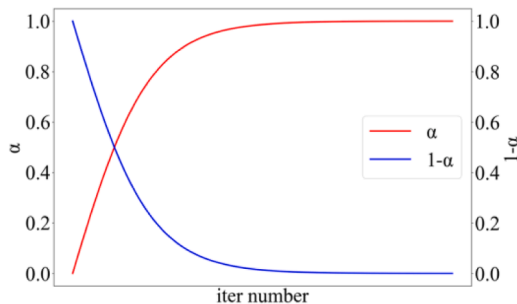


Fig. 3. Dynamic parameter α in the training stage.

Algorithm 1

Source free domain adaptation framework.

Input: unlabeled train dataset $D_t = \{(x_t)|x_t \in X\}_{t=1}^{N_t}$, fixed pre-trained source model (F_s, C_s) , trainable target model (F_t, C_t, C_{s2t}) , number of classes N_{cls} , training epochs Eps , EAPMM update period $update_num$.
Output: Optimized target model (F_t, C_t, C_{s2t})

```

1: Begin
2:  $(F_t, C_t, C_{s2t}) \leftarrow (F_s, C_s)$  // Shared initial parameters
3: for  $c \leftarrow 1$  to  $N_{cls}$  do
4:  $H_c \leftarrow (F_t, C_t, x_t) // Eq.(2)$ 
5:  $\eta \leftarrow \max\{\min(H_c)\}$ 
6: end for
7:  $P_{set1}, P_{set2} = []$  //Initialize EAPMM
8: for  $t \leftarrow 1$  to  $N_t$  do
9:  $P_{c1}, P_{c2} \leftarrow \{(F_t(x_t)|H(x_t) \leq \eta)\}$ 
10:  $P_{set1}.append[P_{c1}], P_{set2}.append[P_{c2}]$  // EAPMM
11: end for
12: for  $c \leftarrow 1$  to  $N_{cls}$  do
13:  $S_c \leftarrow (F_t, P_{c1}, x_t) // Eq.(6)$ 
14:  $\bar{y}_t \leftarrow \arg\max\{S_c(x_t)\}$ 
15: end for
16: for  $e \leftarrow 1$  to  $Eps$  do
17:  $\bar{y}_t \leftarrow (F_s, C_s, x_t)$ 
18:  $c \leftarrow (P_{set1}, P_{set2}, F_t, x_t) // Eq.(10), Eq.(11)&Eq.(12)$ 
19:  $L_{tar}, L_{src} \leftarrow (\bar{y}_t, \bar{y}_t, c, F_t, C_t, C_{s2t}, x_t)$ 
20:  $\alpha \leftarrow (e, Eps) // Eq.(16)$ 
21:  $L_{total} \leftarrow (\alpha, L_{src}, L_{tar}) // Eq.(15)$ 
22: Update:  $(F_t, C_t, C_{s2t})$ 
23: if  $e \% update\_num == 0$  then
24:  $P_{set1}, P_{set2} \leftarrow UpdateP_{set1}, P_{set2} // step3-11$ 
25: end if
26: end for
27: Return  $(F_t, C_t, C_{s2t})$ 

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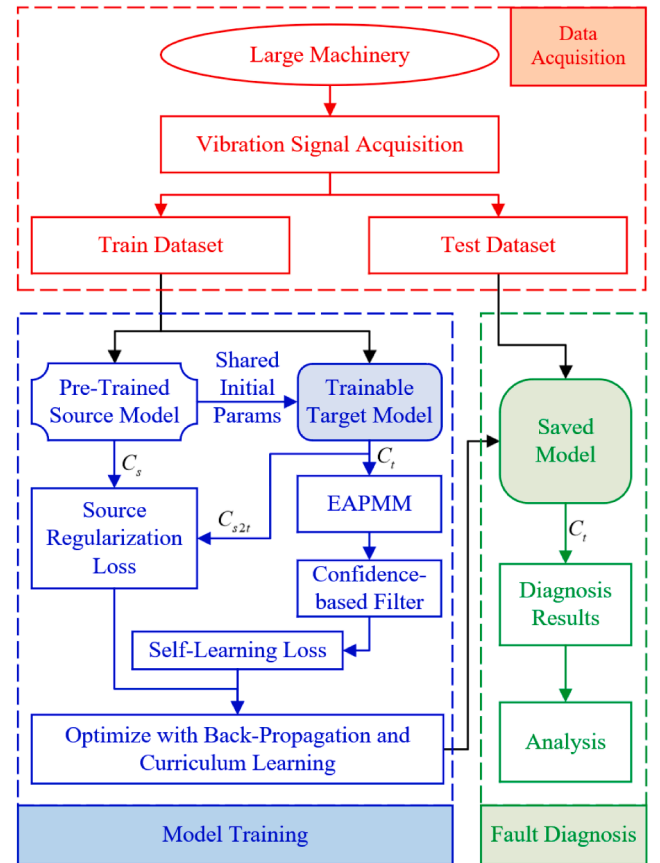


Fig. 4. The fault diagnosis flowchart of the proposed SFDAF.

- (3) Fault Diagnosis: The test dataset is fed into the saved target fault diagnosis model, where the output of the classifier C_t is defined as the diagnosis results, and the results of the fed data are analyzed and evaluated for the health state.

3. Experiments and results

3.1. Compared approaches

In this paper, several popular fault diagnosis approaches were employed for comparison to validate the proposed method. This paper introduces three standards: the conventional DL method (CNN) [33], typically UDA methods, (D-Coral [27], DAN [34], and DANN [35]) and the source-free domain adaptation (SFDA_NCF, without CF compared with the proposed SFDAF). The compared methods are described as follows, and the differences and similarities are listed in Table 1.

This paper mainly focuses on the decouple the domain adaptation with direct usage of the source data through the pre-trained source model. Generally, a deep learning model usually including two parts: feature extractor and classifier. The feature extractor is designed to adaptively learn latent high-dimensional features, and the classifier is employed to map the features to the corresponding label space. The proposed SFDAF does not impose specific requirements on the model structure. However, for a fair comparison, the compared methods should utilize the same structures. In this paper, as the typical deep learning model, CNN is applied as the backbone of the proposed method with a similar structure to the literature [20]. The architecture hyperparameters of the models are shown in Table 2.

3.2. Implementation details

This paper applied the Pytorch framework to build and train all the models, and the backpropagation algorithm optimized the parameters. The hyper-parameters are determined by experimental adjustment. The batch size is 50 for all models, and the maximum epoch number is 20 for SFADA and 60 for compared methods, this is because that the SFDAF proposed in this paper has a faster convergence speed and can reach the optimal effect after 20 epochs, and other method converge at 60 epochs. The initial learning rate is 1E-6 for source free methods and 0.001 for other compared methods, the reason that the lower learning rate of the proposed method is that the optimization of the proposed method is to fine-tune parameters from the pre-trained source model, while other methods optimize from the random initial parameters. The stepLR strategy was applied to adjust the learning rate, optimizing the model in more giant steps at the beginning and smaller stages at the end. The step epoch is 4 for SFADA and 10 for compared methods, which keep the same adjustment number of times for all methods, and the coefficient gamma is 0.1, which means that the learning rate is adjusted to one-tenth of the original for every step trained. In all the methods, the Adam optimizer with a momentum of 0.9 was employed to train models, and the weight decay in L2 regularization is 1E-5.

The central computer hardware used in this paper mainly includes CPU (Intel Core i7-10,700 @ 2.90 GHz), Memory (16GB), and GPU (NVIDIA TITAN XP 12GB).

In this paper, the overall accuracy is utilized to evaluate the classi-

Table 1
Description of compared approaches.

Standard	Method	Domain Adaptation Method	Domain Task
DL	CNN	/	Source data available
UDA	D-Coral	Coral	Source data available
	DAN	MMD	Source data available
	DANN	Adversarial	Source data available
SFDA	SFDA_NCF	EAPMM	Source Free
	SFDAF	EAPMM	Source Free

Table 2
Architecture hyperparameters of the proposed methods.

Module	Layers	parameters
Feature Extractor	Convolution layer-1	$1 \times 17 \times 16$
	MaxPooling layer-1	1×2
	BatchNorm layer-1	1×16
	Convolution layer-2	$1 \times 17 \times 64$
	MaxPooling layer-2	1×2
	BatchNorm layer-2	1×64
	Convolution layer-3	$1 \times 17 \times 128$
	MaxPooling layer-3	1×2
	BatchNorm layer-3	1×128
Classifier	Fully connected layer-1	256
	Fully connected layer-2	64
	Fully connected layer-3	5

fication performance of fault diagnosis models. Table 3 shows the confusion matrix of classification. Accuracy refers to the percentage of correctly diagnosed health states, i.e., the positive condition are diagnosed as positive, and the negative condition are diagnosed as negative. Therefore, the accuracy applied in this paper is defined as follows.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

3.3. Experimental validation

3.3.1. Experimental setup and data description

To verify the effectiveness of the proposed method, the drivetrain transmission diagnostic simulator (DDS) was applied to simulate the fault vibration data. The main component of the DDS is mainly including a driven motor, a planetary gearbox, a fixed axis gearbox, and a magnetic brake. In data acquisition, the NI cDAQ-9188 and a NI 9234 were utilized to obtain the vibration signal with the sample rate of 12,800 Hz, where an accelerometer (PCB 352C03) was applied and mounted on the vertical direction of the planetary gearbox, as shown in Fig. 5.

In this paper, 5 types of gearbox faults were simulated by the DDS, including 2 types of bearing faults (inner race fault and outer race fault), 2 types of gear faults (tooth root crack fault and tooth surface pitting fault) and normal state. The health state description and category labels are shown in Table 4, and the actual pictures of fault gears and fault bearings are shown in Fig. 6.

In this paper, vibration signals in 3 working conditions were simulated with different rotation speeds and torsional loads, as shown in Table 5. To obtain enough data, 4 acquisitions were carried out under each working condition, and each acquisition time was set as 16 s, the time-domain waveforms of singles acquainted under different working conditions are shown in Fig. 7. Then the vibration signals were pre-processed and divided into the training set and test set. Expressly, each sample was 1024 points without overlapping, then each class consists of 800 samples, the ratio of training data to test data was set as 1, and the final number of training samples and test samples in each category was 400. It is worth noting that this paper aiming at the problem of unsupervised source-free domain adaptation, therefore, the labeled source data are unavailable in the training stage of the proposed method.

4. Results and discussion

The detailed results of the SFDAF and the compared approaches under different cross-domain tasks with DDS datasets are provided in

Table 3
The confusion matrix for the classification.

Predicted condition	Positive condition	Negative condition
Positive	True positive (TP)	False positive (FP)
Negative	False negative (FN)	True negative (TN)

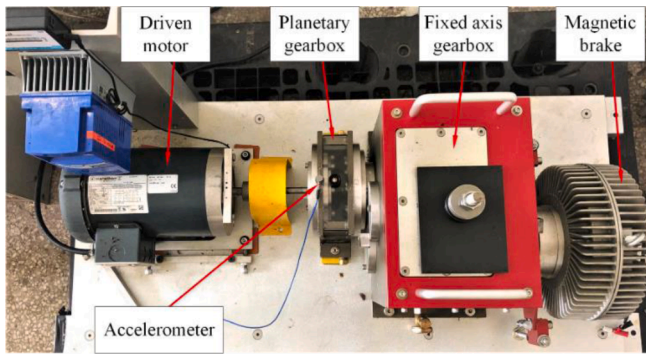


Fig. 5. The experimental device of DDS and the accelerometer location.

Table 4
The Health States of The Planetary Gearbox.

Gearbox health state	Category label
Inner race fault of bearing	IF
Outer race fault of bearing	OF
Normal	N
Tooth root crack fault of gear	CF
Tooth surface pitting fault of gear	PF

Table 6, and ten trials are implemented for each task with the same datasets to get more reliable results.

As can be seen in Table 6, the average accuracy of the SFDAF method proposed in this paper under different cross-domain tasks is 84.11%, which obtained the best performance among all the compared methods. Without the confidence-based filter, the average results of SFDA_NCF is 80.51%, 4.4% lower than the SFDAF. Although there are some pseudo labels are discard, the overall performance is improved, which show the superiority of the confidence-based filter. Results in unsupervised domain adaptation methods exists discrepancies under different tasks, but higher than the results of CNN with 68.72%. Specifically, the average result of the D-Coral in the 6 cross-domain tasks is 83.37%, which obtain the best performance among the conventional domain adaptation methods. Especially in task (B→C), the diagnosis accuracy of the D-Coral is up to 99.09% with a standard deviation of 0.32%, which is 3.67% higher than the proposed method. It is demonstrated that the proposed SFADA can achieve an effect comparable performance to the conventional domain adaptation method without the available source data, which can solve the problem of data privacy and security.

Moreover, there are severe discrepancies among the diagnosis accuracies among 6 cross-domain tasks. Specifically, the average result of the proposed SFDAF in task (B→C) is 95.42%, while there only obtain the 72.34% in task (A→C). The reason for the results bias among different cross-domain tasks is that the vibration signals between different domains are obtained under different working conditions, the

corresponding frequency component and amplitude are highly affected by the rotation speed and loads, which lead to the great distribution discrepancies between different domains.

To verify the convergence of the proposed SFDAF, the accuracy and loss in the training stage were statistically analyzed. Furthermore, the results of the accuracy and loss of the SFDA_NCF were compared to demonstrate the effectiveness of the confidence-based filter. The accuracy curves and the loss curves in the training stage with DDS data under task (C→A) are shown in Fig. 8. The accuracy of the pre-trained model with data domain C is 58.98% under data domain A. This is because that the data discrepancy between domain C and A lead to the poor fit of data distribution of domain A. In the training stage, the model converges in a few epochs of adaptive learning, and the proposed SFDAF obtains a higher accuracy and lower loss than the SFDA_NCF, which shows the outstanding convergence effect of the method and the superiority of the confidence-based filter.

In this paper, the confusion matrix method was applied to further illustrate the fault diagnosis results of the five health states. The confusion matrixes of the CNN, SFDA_NCF and the SFADA under task (C→A) are shown in Fig. 9. The diagonal data indicates the correct number of fault predictions in each category.

As can be seen from Fig. 9(a) that the diagnosis results in classes IF and CF of the CNN are almost incorrect, which mainly be classified as class PF. As for SFDA_NCF, shown in Fig. 9(b), it can be seen that the diagnosis accuracy in classes IF and CF have been significantly improved. Fig. 9(c) shows that the misclassifications in the above two methods have been further reduced, despite the errors in class PF, and obtained the better overall diagnosis accuracy. It is demonstrated that the proposed EAPMM can effectively decouple the fault features in different health state, and the confidence-based filter can further improve the performance of the proposed method.

To further investigate the feature learning ability of the proposed method, a non-linear dimension reduction method, t-distribution Stochastic Neighbor Embedding (t-SNE) was applied to map the high-dimension features of the test data into 2-D spaces.

As shown in Fig. 10(a), the data features of different classes are highly mixed in the original dataset, which cannot be effectively classified. Fig. 10(b) shows the feature distribution of the CNN method, the features of different categories in the target domain can be roughly classified by learning the knowledge of the relevant source domain, while the distribution distance of the classes CF, PF and IF is rather closed, which may lead to the misclassifications. With the proposed

Table 5
Description of working conditions in different domains.

Domain Definition	Rotating Speeds (Hz)	Loads (hp)
Domain A	20	8
Domain B	30	4
Domain C	40	0

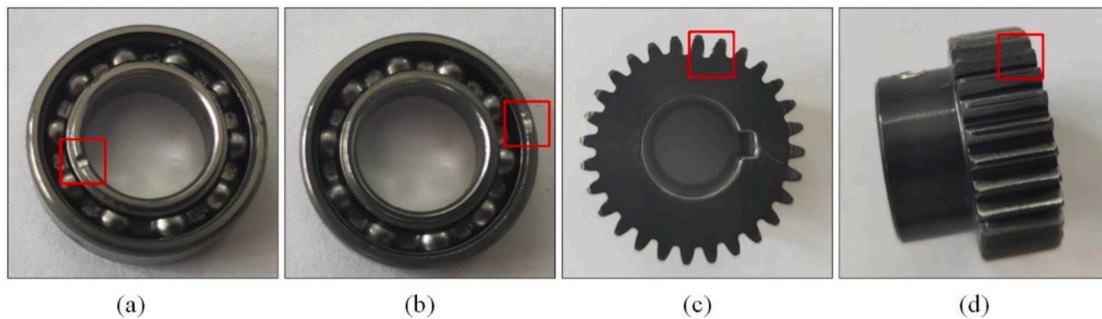


Fig. 6. Fault diagrams of gear and bearing: (a) inner race fault of bearing; (b) outer race fault of bearing; (c) tooth root crack fault of gear; (d) tooth surface pitting fault of gear.

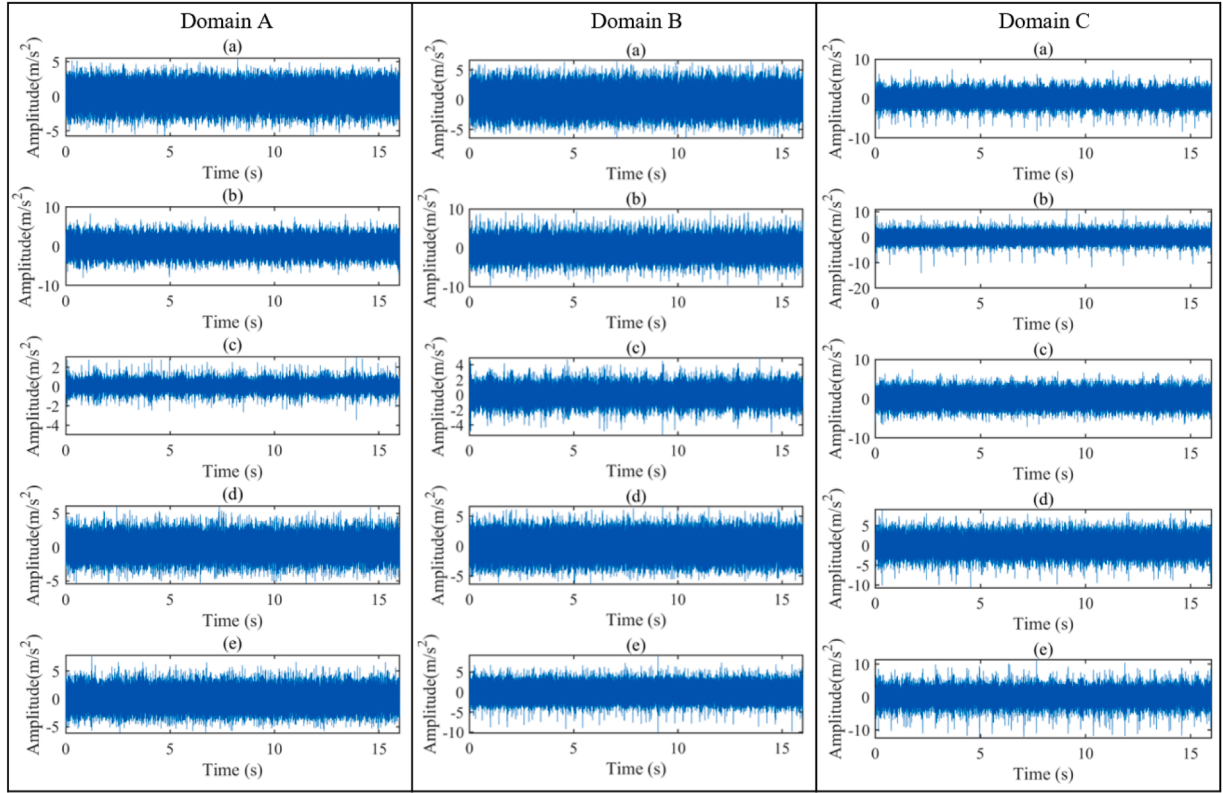


Fig. 7. Raw vibration signals of five health state DDS data in the three domains: (a) inner race fault of bearing; (b) outer race fault of bearing; (c) normal state; (d) tooth root crack fault of gear; (e) tooth surface pitting fault of gear.

Table 6

Classification accuracy with DDS datasets.

Method	Accuracy±Standard Deviation (%)		B→A	B→C	C→A	C→B	Avg
	A→B	A→C					
CNN	71.78±3.80	63.36±2.49	71.11±3.57	89.62±2.62	58.98±2.45	57.45±3.37	68.72
D-CORAL	89.99±4.98	77.80±2.67	82.95±3.55	99.09±0.32	76.31±2.22	74.10±2.05	83.37
DAN	80.61±2.22	76.90±3.78	85.28±3.90	98.85±0.76	72.08±3.15	83.91±3.04	82.93
DANN	84.14±4.13	74.64±3.38	85.29±3.71	96.35±0.62	69.85±3.21	69.34±4.20	79.93
SFDA_NCF	84.65±0.77	71.165±0.15	72.145±0.91	91.59±0.62	80.04±0.95	83.475±1.87	80.51
SFDAF	85.51±0.92	72.34±0.18	81.16±2.10	95.42±0.75	86.28±0.71	84.92±0.24	84.11

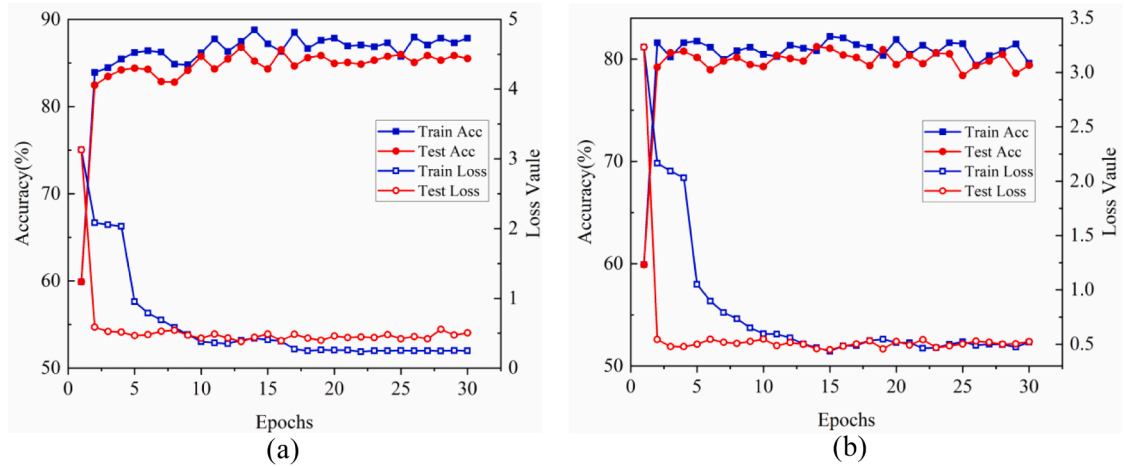


Fig. 8. The accuracy curves and the loss curves in the training stage with DDS data: (a) SFDAF; (b) SFDA_NCF.

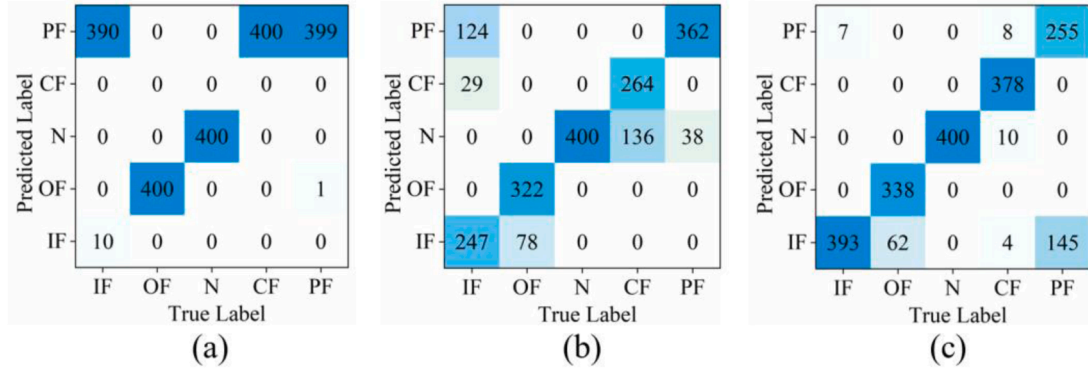


Fig. 9. Confusion Matrix of DDS data: (a) predicted results in CNN; (b) predicted results in SFDA_NCF; (c) predicted results in SFDAF.

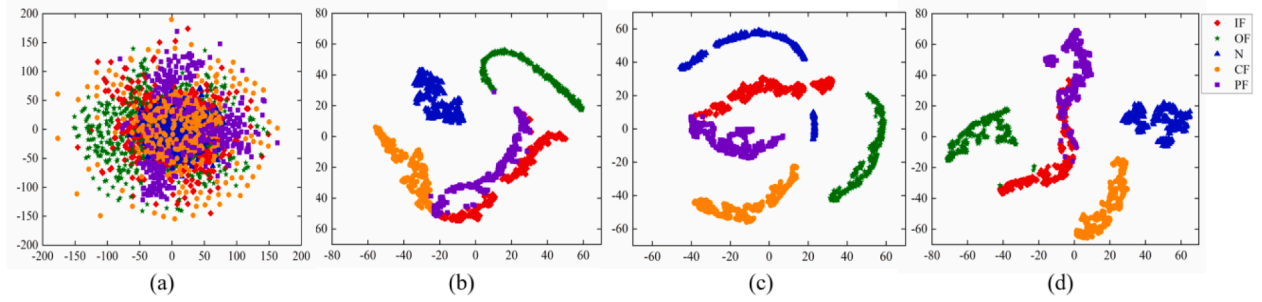


Fig. 10. t-SNE Visualization: (a) original DDS data; (b) features of the CNN; (c) features of the SFDA_NCF; (d) features of the SFDAF.

EAPMM, features in the target domain can be further distinguished in the SFDA_NCF, as shown in Fig. 10(c). Fig. 10(d) represents the feature distribution of the proposed SFDAF, it can be seen that each category has been better distinguished, and the distance between the categories is larger, which obtained the best feature distinction performance. It is demonstrated that the proposed method can help extract more discriminative features, thereby improving the accuracy of fault

diagnosis.

4.1. Engineering applications

4.1.1. Dataset description of wind turbine gearboxes

This section selected several 2 MW wind turbines in a wind farm in China as the application object. The structure of the wind turbine

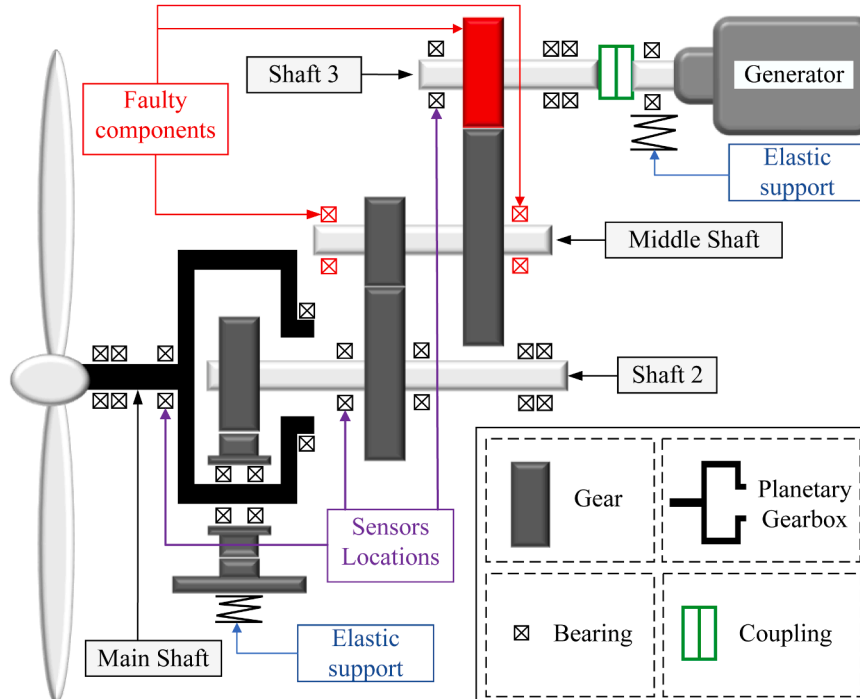


Fig. 11. Schematic diagram of wind turbines transmission system and the locations of sensors.

transmission system mainly includes the main shaft, one-stage planetary gear train, and two-stage parallel gear train, as shown in Fig. 11. The wind farm adopted the Condition Monitoring System (CMS) to collect vibration data every 4 h. Under the long-time accumulated collection, five types of health state vibration data of wind turbine gearboxes were collected and utilized to verify the effectiveness of the proposed method. The detailed description of health states is shown in Table 7.

Due to the environmental influence, the wind speed at the service site of wind turbines changes, leading to the fan blade rotation speed changing with the ambient wind speed. To ensure the reliability and stability of electric power generation, the control system of wind turbines is usually set with limited protection to maintain a reasonable rotation speed range. Therefore, the data collected by the sensor is usually time-varying.

This paper defines different domains by different accelerator sensors in wind turbine gearboxes. As shown in Fig. 11, The sensor locations in the main shaft, shaft 2, and shaft 3 are regarded as domains A, B, and C, respectively, and part of the industrial site is shown in Fig. 12. It is worth noting that the sampling frequency of domain A is 12,800 Hz with the relatively low sampling frequency sensor (WT1351D), and the sampling frequency of domains B and C is 25,600 Hz with the relatively high sampling frequency sensor (AC102-1A), as shown in Table 8.

In the actual data collection process, the sampling time in domain A was set as 15 s, which can obtain 19,200 points in a single acquisition. While the sampling time in domains B and C was set as 5.12 s, and the point in a single acquisition is 131,072. In this paper, training data and test data were divided with the size of 1024 by sliding window without overlapping. Then 1000 samples were obtained in each class, where 400 of them regarded as training set, the other 600 samples were defined as test set. The vibration waveforms of five health states in different domains are shown in Fig. 13.

5. Results and discussion

The detailed results of the SFDAF and the compared approaches under different cross-domain tasks with real wind turbine gearboxes datasets are provided in Table 9, and ten trials are implemented for each task, where the average value was employed as the evaluation index.

As shown in Table 9, due to the interference of changing working conditions and strong noise in the real wind turbine gearboxes data, the average diagnosis results of all the methods are not very satisfactory. However, the proposed method still obtained the best performance under six cross-domain tasks with an accuracy of 67.2%, which is 14.65% higher than that of the CNN. The results of the SFDA NCD is 62.79%, which is comparable to the D-RORAL, DAN and DAN. The above results once again show that the proposed SFDAF can effectively improve the performance of domain adaptation under data privacy and security without the source data.

Furthermore, there is significant bias in the results under different cross-domain tasks. The results in the tasks between domains B and C are rather higher than other tasks, especially in task (B→C) of the DAN, with an accuracy of 87.67%, while only 44.37% in the task (B→A). The main reason for the great bias is that the sampling frequency of domain A is different from domains B and C. Specifically, with the same points, samples with lower sampling frequency contain more cycle information, which can represent more global features. In contrast, the local features

can be better represented in the samples with higher sampling frequency. The difference between the local and global features leads to significant discrepancies which seriously affect the accuracy of the domain adaptation.

The accuracy curves and the loss curves in the training stage with wind data under task(B→C) are shown in Fig. 14. The accuracy of the pre-trained model with data domain B is 74% under data domain C. As can be seen in the Fig. 14, the loss curve and accuracy curve of the SFDA_NCF fluctuates more sharply than that of the SFDAF, this indicates that the quality of the pseudo label is improved with confidence-based filter, which help the model training more stable. Furthermore, in the training stage, the accuracy of the above methods with the wind data is less stable than that of the DDS data. This is because that the actual wind turbine operating environment is complex, the noise interference is serious, and the frequency components of the wind data are complex, while the DDS data is collected in the laboratory environment, the fault characteristics are more obvious, which is conducive to the learning of the model.

The confusion matrix results of the task (B→C) with the CNN, SFDA_NCF and the SFDAF are shown in Fig. 15. As can be seen from Fig. 15(a), the vast majority of category F4 data is wrongly classified as F3 category, and part of the data of the N category is also misjudged as the F2 category. It indicates that the data distribution of F4 and N categories in the target domain is quite different from that in the source domain. Fig. 15(b) shown the results of SFDA_NCF, the misclassification of categories F4 and N have been significantly improved. As shown in Fig. 15(c), despite the misclassification in the categories F4, the general classification in other categories is rather accurate, which indicates the higher diagnosis results in the task. It is demonstrated once again that the proposed SFDAF can effectively decouple the characteristics of different health states of the wind turbine gearbox.

The visualization of the of the dimensionality reduction feature in the domain C is shown in Fig. 16. Fig. 16(a) shows the feature distribution of original data of wind turbine gearboxes, it can be seen that features of the wind turbine gearbox data with different health states are highly mixed and cannot be effectively differentiated. With the adaptive learning by CNN, features in several categories can be classified accurately, but confusion still exists for categories F4 and F3, which is consistent with the results in Fig. 15(a). Fig. 16(c) shows the results of the SFDA_NCF, with the domain adaptation learning, the feature overlap of categories F4 and F3 is alleviated. And the overlap between the F4 and F3 have been further separated in Fig. 16(d), which obtained better discrimination. The results above demonstrated that conventional CNN could learn knowledge through the relevant source domain and applied to the task in target domain, but some data are prone to misclassified due to the data distribution discrepancies. The proposed EAPMM can improve the performance of the cross-domain tasks, and the confidence-based filter can further improve the model to extract more discriminative features, which helps to obtain better performance in new tasks.

5.1. Empirical analysis

The SFDAF proposed in this paper is mainly aimed at the problem of unavailability of source domain data due to data privacy and security. The parameters of the model trained on source data are optimized through unlabeled target data with a self-learning strategy, which make it suitable for target domain tasks. Therefore, an Empirical Analysis is conducted for the proposed method, including learning rate analysis, curriculum learning strategy analysis, and reliable sample analysis.

5.1.1. Learning rate analysis

The learning rate is one of the essential hyperparameters in the gradient descent optimization of deep neural network weight parameters in backpropagation. Its primary function is to set the magnitude of model parameter optimization, which is important for the proposed SFDAF due to the no ground truth available during the self-learning.

Table 7

Health states of the wind turbine gearbox.

Gearbox health state	Category label
Ball wear of the shaft-2 bearing near the smaller gear	F1
Gear fault of the shaft-3	F2
Ball wear of the shaft-2 bearing near the larger gear	F3
Cage fault of the shaft-2 bearing near the larger gear	F4
Normal	N

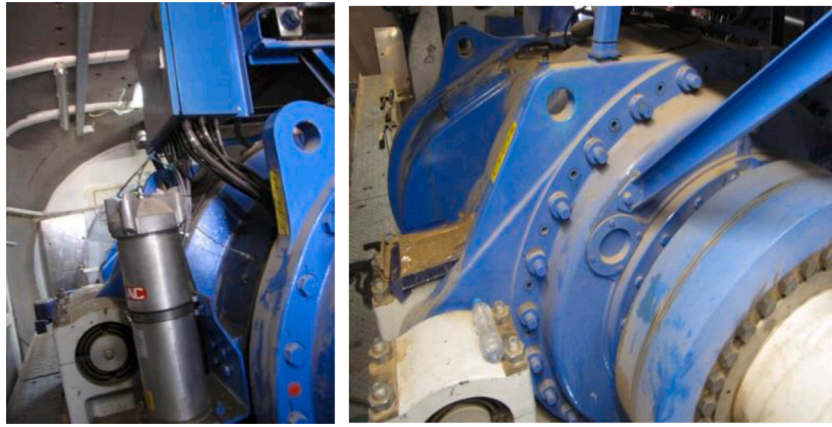


Fig. 12. Part of the industrial site scene.

Table 8

Data description of different domains in wind turbine gearboxes.

Label of Domain	Sampling Frequency	Sensor Location
Domain A	12,800 Hz	Main Shaft
Domain B	25,600 Hz	Shaft 2
Domain C	25,600 Hz	Shaft 3

The domain A data in DDS and the Domain C data of the wind turbine gearboxes are applied in the experimental analysis of the learning rate. The range of the learning rate set from $1E-2$ to $1E-9$, the diagnosis results of the DDS data and wind turbine gearboxes data are shown in Figs. 17 and 18, respectively. As shown in Fig. 17, the initial diagnosis accuracy of the pre-trained model is 60%, and the performance degradation occurred with the larger learning rate ($1E-2$ – $1E-4$), especially

performance collapse with the learning rate of $1E-2$, which means that the model cannot learn effective features. With the decreasing of the learning rate, the accuracy of the SFDAF get increased, where the best performance obtained with the learning rate of $1E-6$. Fig. 17 shows the accuracy of wind turbine gearboxes under different learning rate. Similar to the results of DDS data, a larger learning rate leads to performance degradation while stable accuracy with a lower learning rate, where the best performance still appears in the learning rate of $1E-6$. Therefore, the learning rate of the proposed method in this paper was selected as $1E-6$.

5.1.2. Curriculum learning strategy analysis

The purpose proposed FSDAF is to learn effective features with the unlabeled target domain under the constraint of maintaining the source information. Therefore, it is crucial to balance the model between the

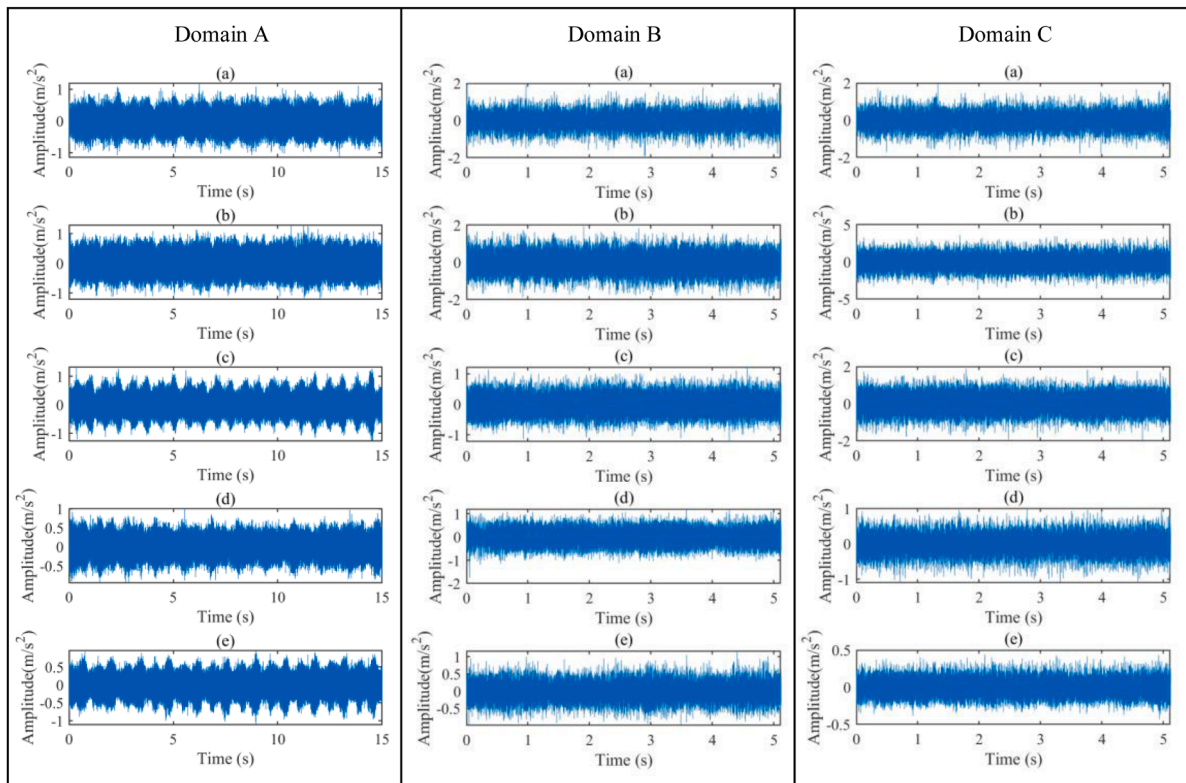


Fig. 13. Vibration Waveform of Five Health State Wind Data in Different Domains: (a) Ball wear of the shaft-2 bearing near the smaller gear; (b) Gear fault of the shaft-3; (c) Ball wear of the shaft-2 bearing near the larger gear; (d) Cage fault of the shaft-2 bearing near the larger gear; (e) Normal.

Table 9
Classification accuracy with wind turbine gearboxes datasets.

Method	Accuracy±Standard Deviation (%)						avg
	A→B	A→C	B→A	B→C	C→A	C→B	
CNN	49.62 ±3.82	42.67 ±5.51	41.62 ±3.95	71.01 ±3.98	47.11 ±4.87	63.26 ±2.45	52.55
D-CORAL	56.34 ±3.75	63.15 ±3.19	50.6 ±4.58	81.83 ±2.63	51.25 ±3.09	70.81 ±5.15	64.35
DAN	57.87 ±3.74	63.02 ±4.85	44.37 ±4.29	87.67 ±3.07	55.26 ±3.83	77.93 ±2.81	62.33
DANN	63.12 ±4.61	66.99 ±4.4	48.81 ±6.27	86.99 ±3.87	55.47 ±4.85	81.07 ±1.38	67.08
SFDA_NCF	59.80 ±1.21	60.10 ±0.18	52.15 ±0.16	80.20 ±0.23	56.95 ±0.15	67.55 ±0.17	62.79
SFDAF	63.41 ±1.09	61.73 ±0.65	61.03 ±2.05	83.96 ±1.24	57.65 ±1.52	75.42 ±0.94	67.20

source domain constraint and the target domain learning.

To validate the effectiveness of the curriculum learning strategy, experiments with different static α are set to compare that with the dynamic parameter. Figs. 19 and 20 indicated the performance change of the proposed SFDAF with different parameters α under domain A in DDS data and domain C in wind turbine gearboxes data. As shown in Fig. 19, the highest accuracy of the DDS data appears with the α set as 0.6. From Fig. 20, the best performance for wind turbine gearbox data was achieved when the α set as 0.2. It is worth noting that the performance with the curriculum learning strategy is superior to the static setting of the α . It is demonstrated that the curriculum learning strategy can provide a better parameter optimization direction for the model, which balance the model between retaining the source domain information and learning the target domain information.

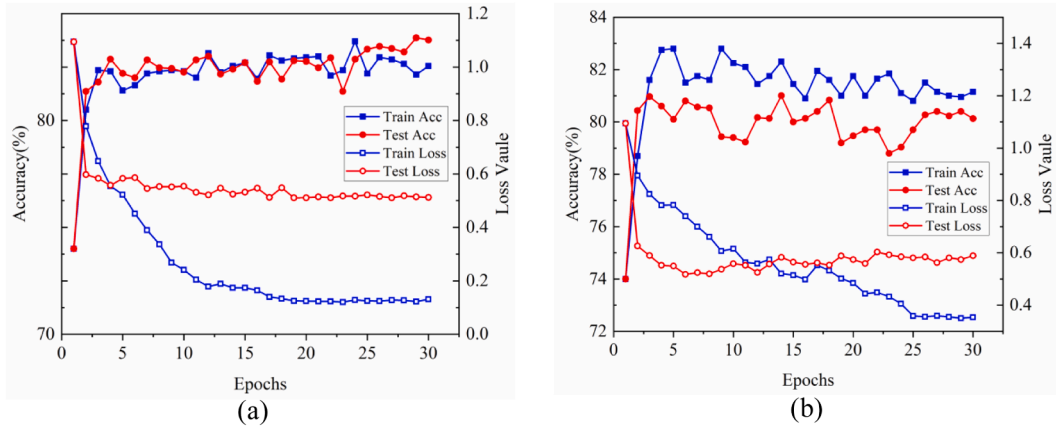


Fig. 14. The accuracy curves and the loss curves in the training stage with wind data: (a) SFDAF; (b) SFDA_NCF.

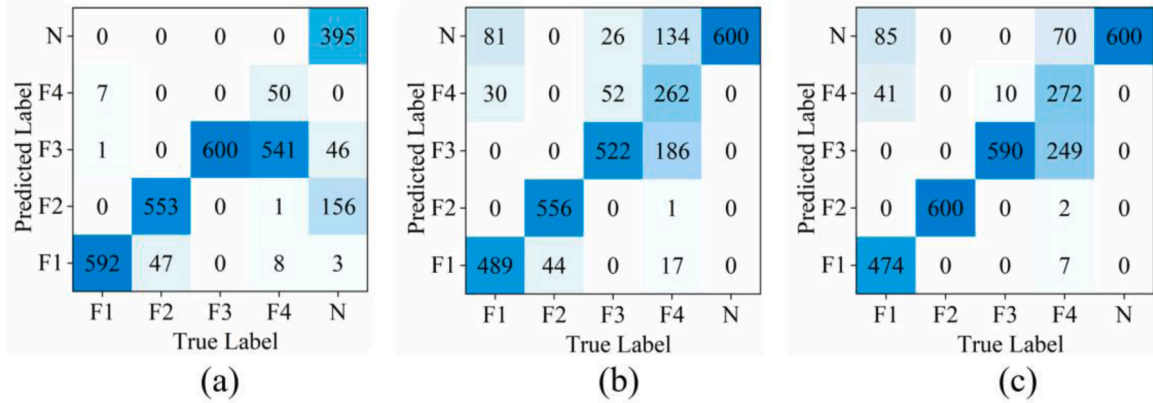


Fig. 15. Confusion Matrix of wind data: (a) predicted results in CNN; (b) predicted results in SFDA_NCF; (c) predicted results in SFDAF.

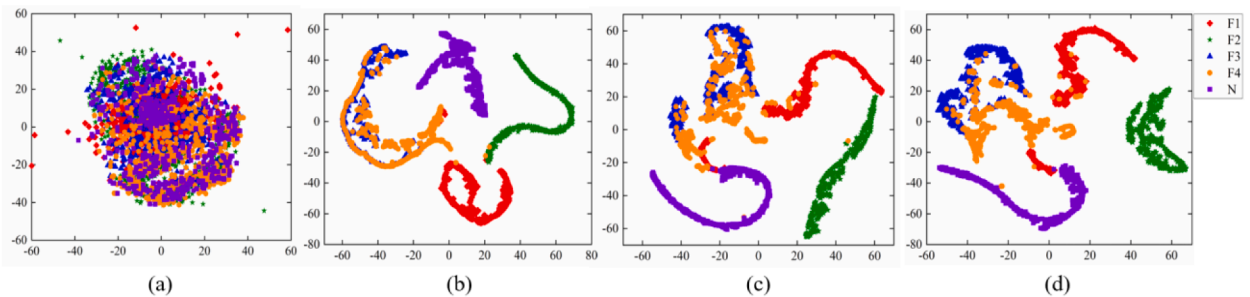


Fig. 16. Visualization of the wind turbine gearbox data: (a) original data; (b) output of the CNN; (c) output of the SFDA_NCF; (d) output of the SFDA.

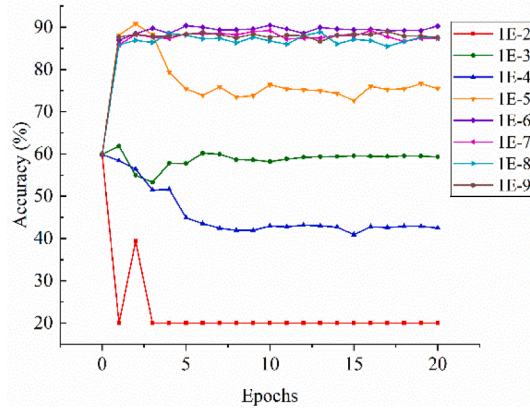


Fig. 17. Test accuracy of domain A in DDS data with different learning rate.

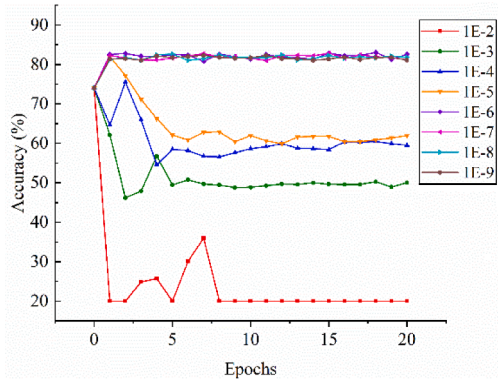


Fig. 18. Test accuracy of the domain C in wind turbine gearboxes data with different learning rate.

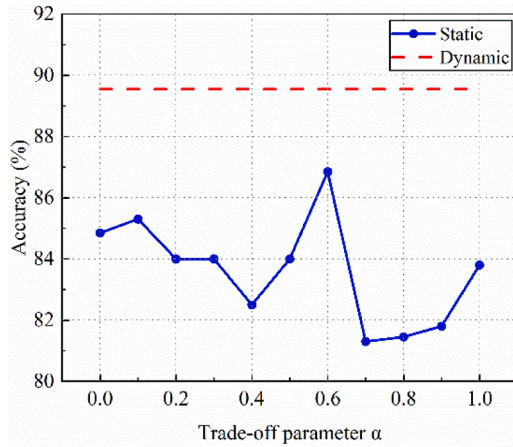


Fig. 19. Performance change of DDS data with different parameter α .

5.1.3. Reliable samples analysis

To further analyze the performance of the proposed self-learning manner, a statistical analysis experiment of the reliable samples in different categories is conducted during the training stage. In the training stage, the best performance was obtained when the EAPMM updated every 4 epochs, therefore, the update period of EAPMM is set as 4 epochs in this section, and then analyze the number of reliable samples with increasing training epochs.

Figs. 21 and 22 indicated the reliable samples between classes in the training stage under domain A in DDS data and domain C in wind

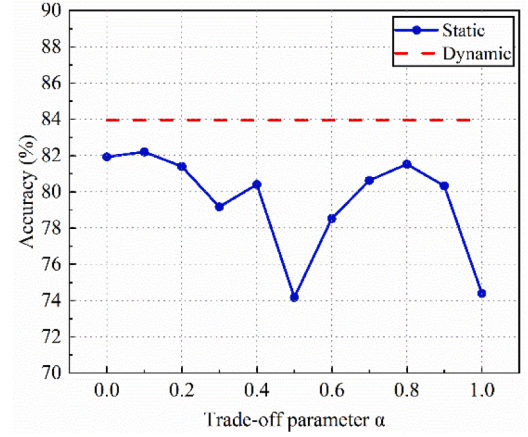


Fig. 20. Performance change of wind data with different parameters α .

turbine gearboxes data. It can be seen from Fig. 21, the number of reliable samples between categories is rather deviated, which varies drastically at the first update and then gradually stabilized. Fig. 22 shows similar results to Fig. 21, starting with the significant deviation and then stabilizing after the first update, which indicates the superiority of the proposed SFDAF regarding the fast convergence. The reason for the significant deviation of reliable sample number among different categories is the distribution discrepancies among the classes, which leads to the large discrepancies in the self-entropy of different classes. The variations of reliable samples demonstrated the effectiveness of the proposed EAPMM.

6. Conclusions

This paper presents a source-free domain adaptation framework (SFDAF) for fault diagnosis of rotation machinery. The proposed method addresses the problem of unavailable source data in domain adaptation by decoupling the direct usage of the source data through the pre-trained source model. Specifically, an adaptive entropy thresholds method is proposed without additional hyperparameters, which construct the entropy-based adaptive prototype memory matrix to obtain the pseudo target labels, then, the quality of the pseudo target labels is further improved by the confidence-based filter, furthermore, a training strategy based on curriculum learning is proposed to help gradually from “easy” to “hard” transfer the focus of the pre-trained model from retaining source domain information to learning target features. Extensive experiments on the datasets of the DDS test rig and real wind turbine gearboxes are conducted, which demonstrated that the

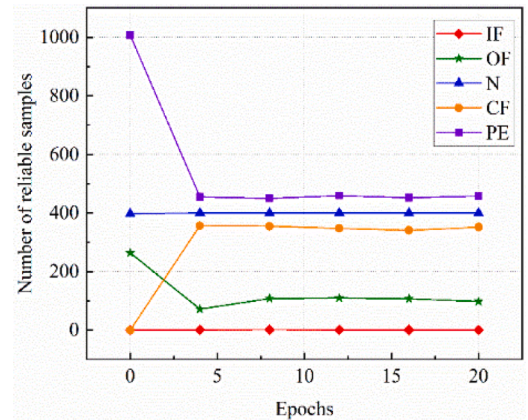


Fig. 21. Reliable samples between different classes with DDS in the training stage under domain A.

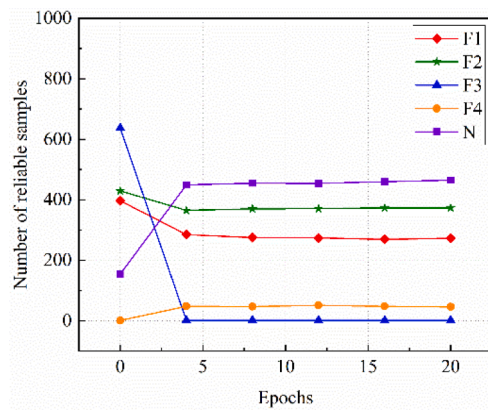


Fig. 22. Reliable samples between different classes with wind data in the training stage under domain C.

proposed SFDAF can effectively improve the performance of domain adaptation without the source data and have achieved the comparable performance compared to typical domain adaptation methods, such as DANN, DAN, D-CORAL. Furthermore, the method presented in this article can accurately diagnose equipment failures, providing strong support for ensuring safety in production and improving equipment reliability as well as data security. In the future, open-set fault diagnosis based on SFDAF is worth studying.

CRedit authorship contribution statement

Qikang Li: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Baoping Tang:** Supervision, Project administration, Funding acquisition, Writing – review & editing. **Lei Deng:** Writing – review & editing, Formal analysis. **Peng Zhu:** Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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